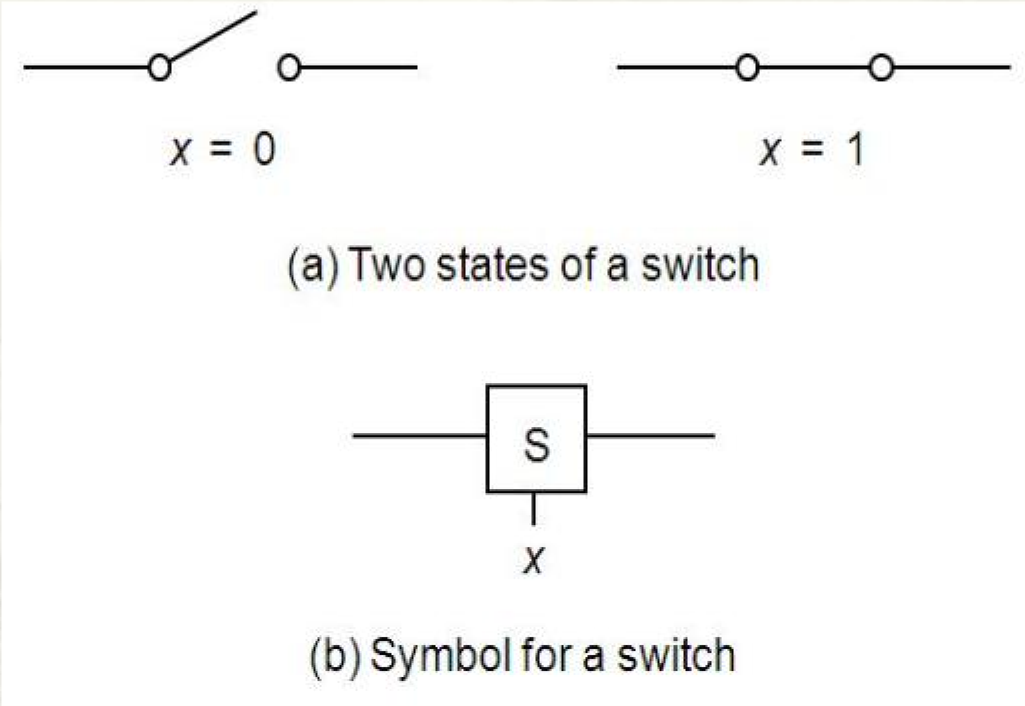
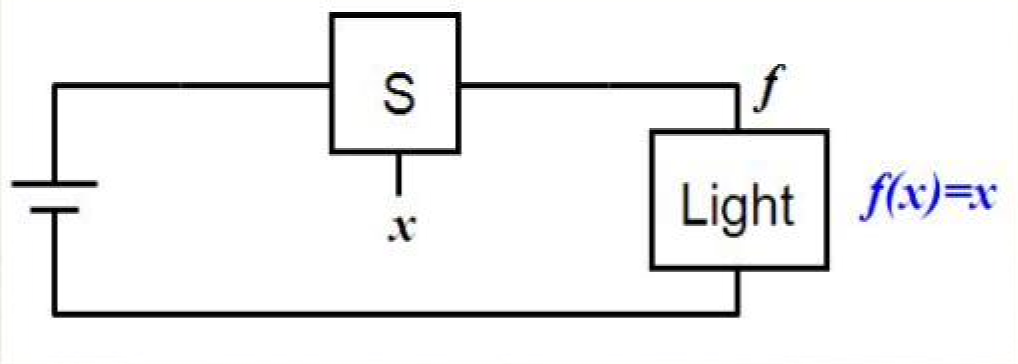
CH2 Fundamental 2:
Boolean Algebra

2.1 Variables and Functions

2.1.1 Switch Algebra



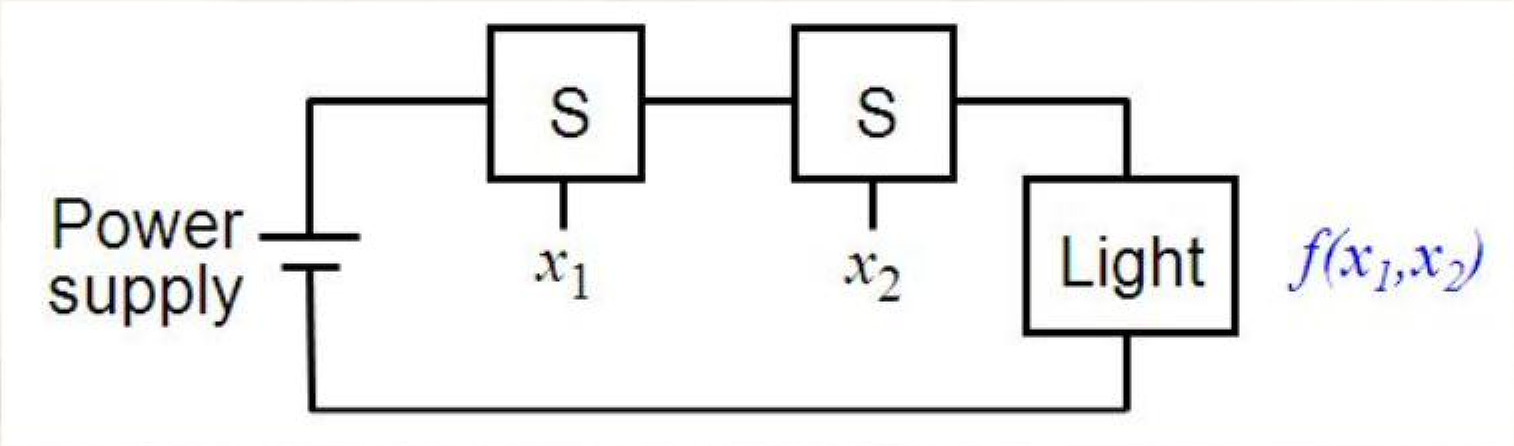


A simple application

2.1.2 Basic Logic Functions(Operations)

- **AND** (logical product)
- **OR** (logical sum)
- **NOT** (Logical complement)

1. AND function



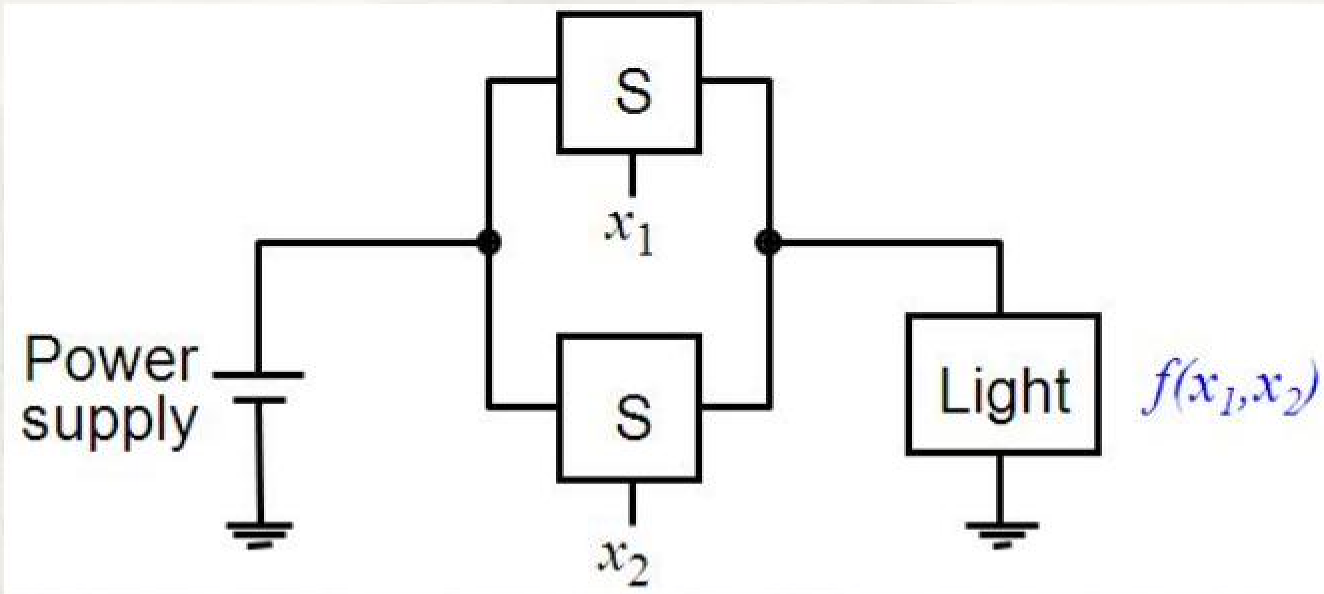
$$f(x_1, x_2) = x_1 \cdot x_2 = x_1 x_2$$

f=1 only if x_1=1 AND x_2=1, otherwise f=0

Logical Product and product term

Note the difference with multiplication

2. OR function



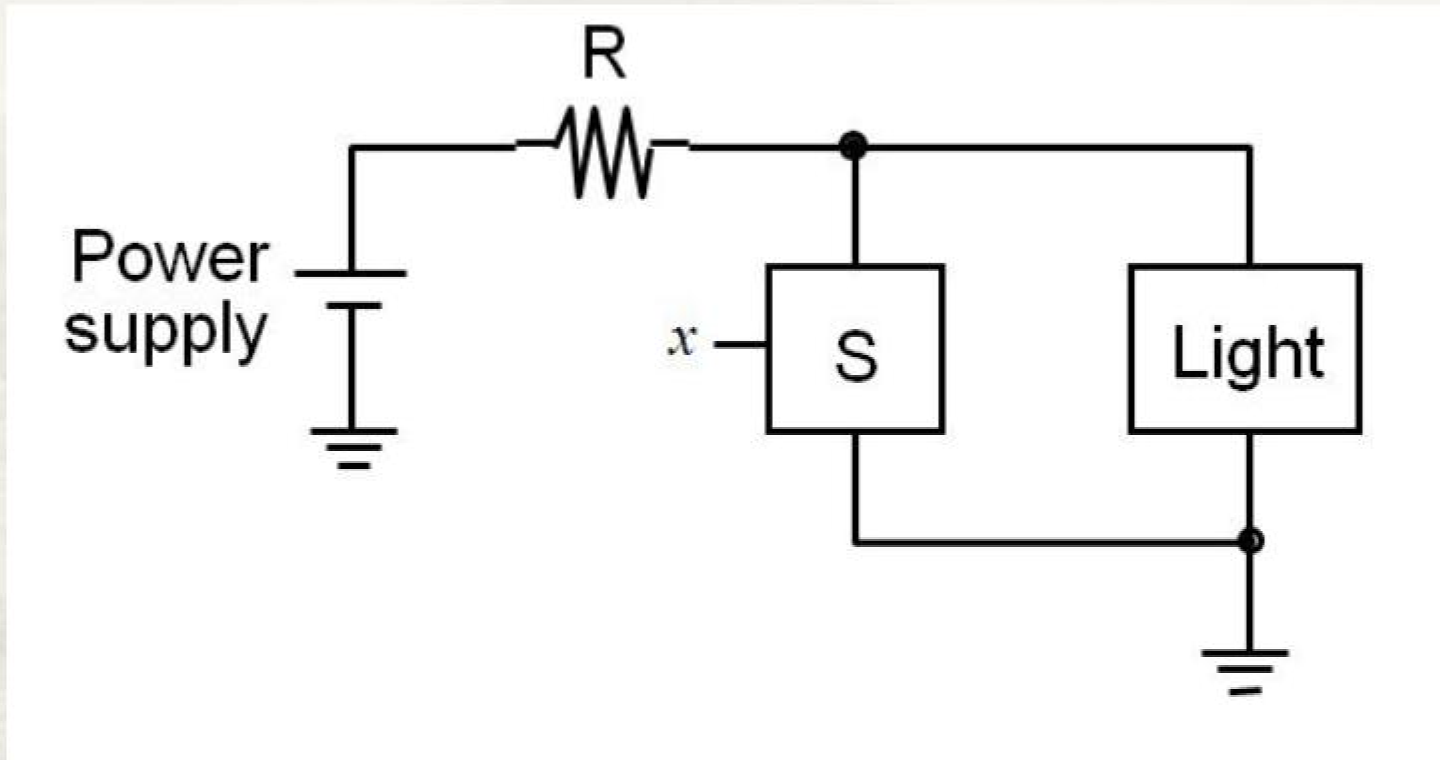
f=1 if x_1=1 OR x_2=1,

$$f(x_1, x_2) = x_1 + x_2$$

Logical sum and sum term

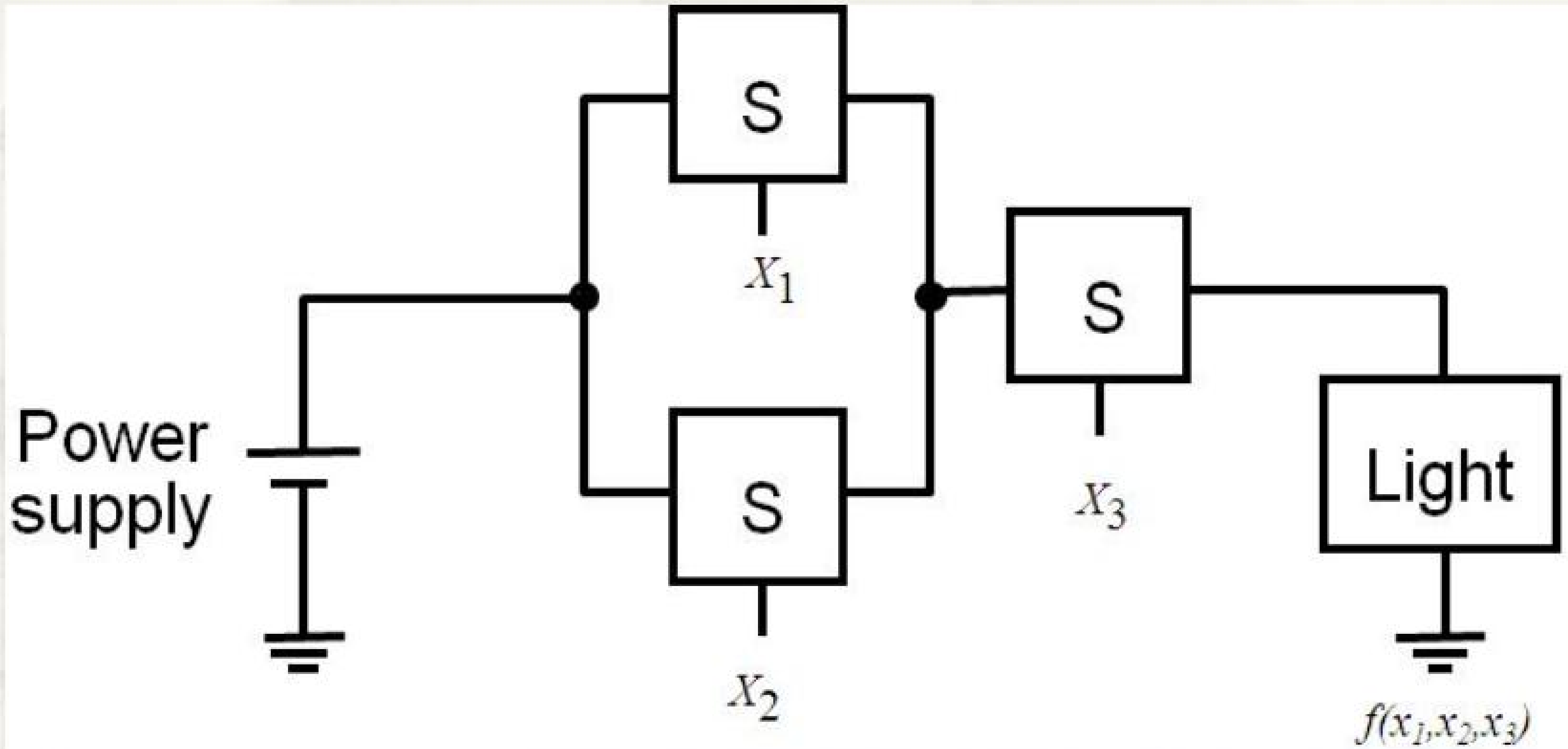
Note the difference from addition

3. NOT function



$$f(x) = \bar{x} = NOT\ x = !x = \sim x = x'$$

4. E.g. of Composite Functions



$$f(x_1, x_2, x_3) = (x_1 + x_2)x_3$$

2.2 Truth table for Logic Functions

x_1	x_2	$x_1 \cdot x_2$	$x_1 + x_2$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1
		AND	OR

2-input AND and OR operations.

x_1	x_2	x_3	$x_1 \cdot x_2 \cdot x_3$	$x_1 + x_2 + x_3$
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

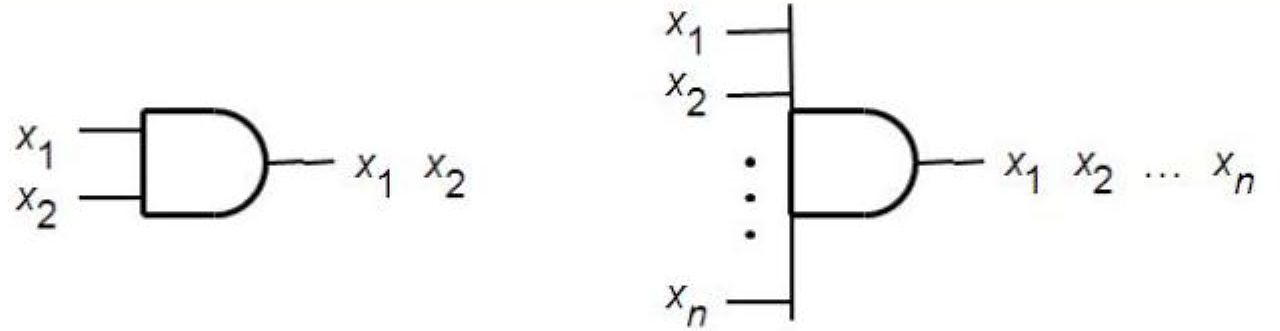
3-input AND and OR operations.

Questions:

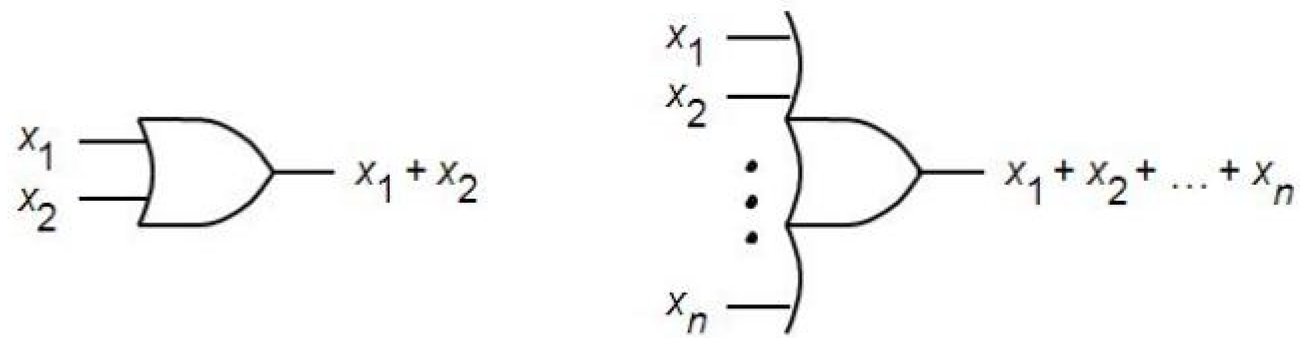
- Truth table for NOT function?
- Truth table for XOR function?
- Truth table for $f(x_1, x_2, x_3) = (x_1 + x_2)x_3$?

X1	X2	X3	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

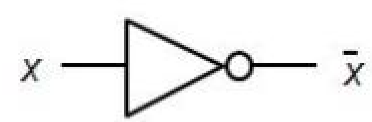
2.3 Basic Logic Gates



(a) AND gates



(b) OR gates



(c) NOT gate

2.4 Rules in Boolean Algebra

2.4.1 Axioms

$$1a.\ \ 0 \cdot 0 = 0$$

$$1b.\ \ 1 + 1 = 1$$

$$2a.\ \ 1 \cdot 1 = 1$$

$$2b.\ \ 0 + 0 = 0$$

$$3a.\ \ 0 \cdot 1 = 1 \cdot 0 = 0$$

$$3b.\ \ 1 + 0 = 0 + 1 = 1$$

$$4a.\ \ \text{if}\ \ x = 0 \Rightarrow \overline{\mathrm{x}} = 1$$

$$4b.\ \ \text{if}\ \ x = 1 \Rightarrow \overline{\mathrm{x}} = 0$$

2.4.2 Single-Variable Theorems

$$5a.\ x \cdot 0 = 0$$

$$5b.\ x + 1 = 1$$

$$6a.\ x \cdot 1 = x$$

$$6b.\ x + 0 = x$$

$$7a.\ x \cdot x = x$$

$$7b.\ x + x = x$$

$$8a.\ x \cdot \overline{x} = 0$$

$$8b.\ x + \overline{x} = 1$$

$$9.\ \overline{\overline{x}} = x$$

2.4.3 2- and 3-Variable ones

10a.	$xy = yx$	*Commutative*
10b.	$x + y = y + x$	
11a.	$x(yz) = (xy)z$	*Associative*
11b.	$x + (y + z) = (x + y) + z$	
12a.	$x(y + z) = xy + xz$	*Distributive*
12b.	$x + yz = (x + y)(x + z)$	
13a.	$x + xy = x$	*Absorption*
13b.	$x(x + y) = x$	
14a.	$x + \overline{x}y = x + y$	
14b.	$x(\overline{x} + y) = xy$	

2.4.3 2- and 3-Variable Identities

15a.	$xy + x\overline{y} = x$	*Combining*
15b.	$(x + y)(x + \overline{y}) = x$	
16a.	$\overline{xy} = \overline{x} + \overline{y}$	*DeMorgan's theorem*
16b.	$\overline{x + y} = \overline{x} \cdot \overline{y}$	
17a.	$xy + \overline{x}z + yz = xy + \overline{x}z$	*Consensus*
17b.	$(x + y)(\overline{x} + z)(y + z) = (x + y)(\overline{x} + z)$	

Simple applications

1. $A\overline{B} + B + BCD$

$= A + B + BCD$ *Absorption*

$= A + B$ *Absorption*

2. $AB + \overline{A} \cdot \overline{B}C + BC$

$= AB + C(\overline{A} \cdot \overline{B} + B)$ *Distributive and Commutative*

$= AB + C(\overline{A} + B)$ *Absorption*

$= AB + \overline{A}C + BC$ *Distributive*

$= AB + \overline{A}C$ *Consensus*

2.4.4 Other theorems

1. DeMorgan's theorem

$$(1)\quad \overline{x_1+x_2+\ldots+x_n}=\overline{x_1}\cdot\overline{x_2}\cdot\ldots\cdot\overline{x_n}$$

$$(2)\quad \overline{x_1\cdot x_2\cdot\ldots\cdot x_n}=\overline{x_1}+\overline{x_2}+\ldots+\overline{x_n}$$

2. Theorem of Duality

$$\mathit{If}\ \ F = f(x_1, x_2, \ldots, x_n, 0, 1, +, \cdot)$$
$$\mathit{then}\ F_d = f(x_1, x_2, \ldots, x_n, 1, 0, \cdot, +)$$

The order of operations must be UNCHANGED!

e.g.:

$F=\overline{A}B+A\overline{B}\overline{C}$ $\qquad$ $F_d=(\overline{A}+B)(A+\overline{B}+\overline{C})$

$F=A(\overline{B}+CD)+E$ $\qquad$ $F_d=[A+\overline{B}(C+D)]E$

$F=(A+0)(B+C\cdot 1)$ $\qquad$ $F_d=A\cdot 1+B(C+0)$

$F=A+\overline{B+\overline{\overline{C}+\overline{D+\overline{E}}}}$ $\qquad$ $F_d=A\cdot\overline{B\cdot\overline{\overline{C}\cdot\overline{D\cdot\overline{E}}}}$

Exercise: find the dual of $F=\overline{A}B+A\cdot\overline{B}\cdot(C+\overline{D})$

$F_d=(\overline{A}+B)\cdot(A+\overline{B}+C\overline{D})$

3. Theorem of inverse functions

$$If\ \ F = f(x_1, x_2, \ldots, x_n, 0, 1, +, \cdot)$$

$$then\ \overline{F} = f(\overline{x_1}, \overline{x_2}, \ldots, \overline{x_n}, 1, 0, \cdot, +)$$

e.g.:

$F(A,B,C,D) = \overline{A} + B(\overline{C} + D)$

$F = \overline{A}B + A\overline{B}\,\overline{C}$

$\overline{F} = (A+\overline{B})(\overline{A}+B+C)$

$F = A(\overline{B}+CD) + E$

$\overline{F} = [\overline{A}+B(\overline{C}+\overline{D})]\overline{E}$

$F = (A+0)(B+C\cdot 1)$

$\overline{F} = \overline{A}\cdot 1 + \overline{B}(\overline{C}+0)$

$F = A+\overline{B+\overline{\overline{C}+\overline{D+\overline{E}}}}$

$\overline{F} = \overline{A}\cdot\overline{\overline{B}\cdot\overline{C\cdot\overline{\overline{D}\cdot E}}}$

Exercise: find the inverse of $F = \overline{A}B + \underline{A\cdot\overline{B}\cdot(C+\overline{D})}$

$\overline{F} = (A+\overline{B})(\overline{A}+B+\overline{C}D)$

2.5 General Forms of Logic functions:
Sum-of-Products(SOP) and Product-of-Sums(POS)
2.5.1 SOP: AND-OR

- A function in form of the truth table

x_1	x_2	$f(x_1, x_2)$	
0	0	1	$\overline{x1}\,\overline{x2}$
0	1	1	$\overline{x1}x2$
1	0	0	
1	1	1	$x1x2$

$$f(x_1, x_2) = \overline{x1}\,\overline{x2} + \overline{x1}x2 + x1x2$$

Minterms

- A minterm is a product term in which each of variables appears once and only once, either in uncomplemented or complemented form.

x_1	x_2	$f(x_1, x_2)$	
0	0	1	$\overline{x1}\,\overline{x2}$
0	1	1	$\overline{x1}x2$
1	0	0	
1	1	1	$x1x2$

Simplified representation for minterms

- m_i: i is determined by:
  - Fixing the order of variables
  - Writing 0 for $\overline{x}$
  - Writing 1 for x
  - i is the decimal value of the sequence.

x_1	x_2	$f(x_1, x_2)$		
0	0	1	$\overline{x1}\,\overline{x2}$	m_0
0	1	1	$\overline{x1}x2$	m_1
1	0	0		
1	1	1	$x1x2$	m_3

$$f(x_1, x_2) = m_0 + m_1 + m_3 = \sum m(0,1,3)$$

Canonical SOP form

'i' is also determined by the row number!

Canonical SOP

- The canonical SOP form consists of only minterms.
- How to find the canonical SOP given an arbitrary expression?

$$F(A, B, C) = \bar{A}B + A\bar{B}\bar{C} + BC$$

- E.g.:

What's more…

x_1	x_2	$f(x_1, x_2)$		
0	0	1	$\overline{x1}\,\overline{x2}$	m_0
0	1	1	$\overline{x1}x2$	m_1
1	0	0		
1	1	1	$x1x2$	m_3

$$f(x_1, x_2) = \overline{x1}\,\overline{x2} + \overline{x1}x2 + x1x2$$ **Canonical SOP**

$$= \overline{x1}(\overline{x2} + x2) + x1x2$$ **Non-SOP**

$$= \overline{x1} + x1x2$$ **SOP**

$$= \overline{x1} + x2$$ **Minimum-cost SOP**

Another example

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$f(x_1, x_2, x_3) = \overline{x1}\,\overline{x2}x3 + x1\overline{x2}\,\overline{x3} + x1\overline{x2}x3 + x1x2\overline{x3}$$

$$= \sum m(1, 4, 5, 6)$$

Three-variable minterms

Row number	x_1	x_2	x_3	Minterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2 x_3$
2	0	1	0	$m_2 = \overline{x}_1 x_2\overline{x}_3$
3	0	1	1	$m_3 = \overline{x}_1 x_2 x_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$
5	1	0	1	$m_5 = x_1\overline{x}_2 x_3$
6	1	1	0	$m_6 = x_1 x_2\overline{x}_3$
7	1	1	1	$m_7 = x_1 x_2 x_3$

2.5.2 POS: OR-AND

- A function in the form of truth table

x_1	x_2	$f(x_1, x_2)$	
0	0	1	$\overline{x1}\,\overline{x2}$
0	1	1	$\overline{x1}x2$
1	0	0	$\overline{m2} = \overline{x1\overline{x2}} = \overline{x1} + x2$
1	1	1	$x1x2$

$$\because \overline{f} = m_2$$

$$\therefore f = \overline{m_2} = \overline{x1\overline{x2}} = (\overline{x1} + x2)$$

Maxterms

- A maxterm is a sum term in which each of variables appears once and only once, either in uncomplemented or complemented form.
- M_i : i is determined by:
  - Fix the order of variables
  - Write 1 for $\bar{x}$
  - Write 0 for x
  - i is the decimal value of the sequence.
- i is also indicated by the row number.

i is also indicated by the row number.

x_1	x_2	$f(x_1, x_2)$	
0	0	1	
0	1	1	
1	0	0	$\leftarrow M_2$
1	1	1	

$$f = \overline{m_2} = \overline{x1\overline{x2}} = \overline{x1} + x2 = M_2$$

Another example

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$\because \overline{f}(x_1, x_2, x_3) = m_0 + m_2 + m_3 + m_7$$

$$\begin{aligned}\therefore f(x_1, x_2, x_3) &= \overline{m_0 + m_2 + m_3 + m_7} \\ &= \overline{m_0} \cdot \overline{m_2} \cdot \overline{m_3} \cdot \overline{m_7} \\ &= M_0 \cdot M_2 \cdot M_3 \cdot M_7 \\ &= (x1 + x2 + x3)(x1 + \overline{x2} + x3)(x1 + \overline{x2} + \overline{x3})(\overline{x1} + \overline{x2} + \overline{x3}) \\ &= \prod M(0, 2, 3, 7)\end{aligned}$$

Canonical POS

Canonical POS form

- The POS form consists of only maxterms.

What's more…

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$f(x_1, x_2, x_3) = \prod M(0, 2, 3, 7)$$
$$= \ldots$$
$$= (x1 + x3)(\overline{x2} + \overline{x3})$$

Minimum-cost POS

Given $F(A, B, C) = \bar{A}B + A\bar{B}\bar{C} + BC$ **Find its canonical POS.**

$$
\begin{aligned}
F(A, B, C) &= (\bar{A}B+A)\ (\bar{A}B+\bar{B})\ (\bar{A}B+\bar{C}) + BC \\
&= (\bar{A}+A)\ (B+A)\ (\bar{A}+\bar{B})\ (B+\bar{B})\ (\bar{A}+\bar{C})\ (B+\bar{C}) + BC \\
&= (A+B)\ (\bar{A}+\bar{B})\ (\bar{A}+\bar{C})\ (B+\bar{C}) + BC \\
&= ((A+B)\ (\bar{A}+\bar{B})\ (\bar{A}+\bar{C})\ (B+\bar{C}) + B) \\
&\quad \cdot ((A+B)\ (\bar{A}+\bar{B})\ (\bar{A}+\bar{C})\ (B+\bar{C}) + C) \\
&= ((A+B+B)\ (\bar{A}+\bar{B}+B)\ (\bar{A}+\bar{C}+B)\ (B+\bar{C}+B)) \\
&\quad \cdot ((A+B+C)\ (\bar{A}+\bar{B}+C)\ (\bar{A}+\bar{C}+C)\ (B+\bar{C}+C)) \\
&= (A+B)\ (\bar{A}+B+\bar{C})\ (B+\bar{C})\ (A+B+C)\ (\bar{A}+\bar{B}+C) \\
&= (A+B+C\bar{C})\ (\bar{A}+B+\bar{C})\ (A\bar{A}+B+\bar{C})\ (A+B+C)\ (\bar{A}+\bar{B}+C) \\
&= (A+B+C)\ (A+B+\bar{C})\ (\bar{A}+B+\bar{C})\ (\bar{A}+\bar{B}+C)
\end{aligned}
$$

$$
\begin{array}{cccc}
\downarrow\downarrow\downarrow & \downarrow\downarrow\downarrow & \downarrow\downarrow\downarrow & \downarrow\downarrow\downarrow \\
0\ 0\ 0 & 0\ 0\ 1 & 1\ 0\ 1 & 1\ 1\ 0
\end{array}
$$

$$= \prod M(0, 1, 5, 6)$$

What do you think of the algebraic approach?

Review previous examples and seek a shortcut

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$\because \overline{f}(x_1, x_2, x_3) = m_0 + m_2 + m_3 + m_7$$

$$\begin{aligned}
\therefore f(x_1, x_2, x_3) &= \overline{m_0 + m_2 + m_3 + m_7} \\
&= \overline{m_0} \cdot \overline{m_2} \cdot \overline{m_3} \cdot \overline{m_7} \\
&= M_0 \cdot M_2 \cdot M_3 \cdot M_7 \\
&= (x1 + x2 + x3)(x1 + \overline{x2} + x3)(x1 + \overline{x2} + \overline{x3})(\overline{x1} + \overline{x2} + \overline{x3}) \\
&= \prod M(0, 2, 3, 7)
\end{aligned}$$

Canonical POS

Questions:

- Relationship between canonical SOP and canonical POS?

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$\begin{aligned} f(x_1, x_2, x_3) &= \sum m(1,4,5,6) \\ &= \prod M(0,2,3,7) \end{aligned}$$

Three-variable minterms and maxterms.

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2x_3$	$M_1 = x_1 + x_2 + \overline{x}_3$
2	0	1	0	$m_2 = \overline{x}_1x_2\overline{x}_3$	$M_2 = x_1 + \overline{x}_2 + x_3$
3	0	1	1	$m_3 = \overline{x}_1x_2x_3$	$M_3 = x_1 + \overline{x}_2 + \overline{x}_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$	$M_4 = \overline{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\overline{x}_2x_3$	$M_5 = \overline{x}_1 + x_2 + \overline{x}_3$
6	1	1	0	$m_6 = x_1x_2\overline{x}_3$	$M_6 = \overline{x}_1 + \overline{x}_2 + x_3$
7	1	1	1	$m_7 = x_1x_2x_3$	$M_7 = \overline{x}_1 + \overline{x}_2 + \overline{x}_3$

2.6 Finding the minimum-cost functions: algebraic approach

13a.	$x + xy = x$	*Absorption*
13b.	$x(x + y) = x$	
14a.	$x + \overline{x}y = x + y$	
14b.	$x(\overline{x} + y) = xy$	
15a.	$xy + x\overline{y} = x$	*Combining*
15b.	$(x + y)(x + \overline{y}) = x$	
16a.	$\overline{xy} = \overline{x} + \overline{y}$	*DeMorgan's theorem*
16b.	$\overline{x + y} = \overline{x} \cdot \overline{y}$	
17a.	$xy + \overline{x}z + yz = xy + \overline{x}z$	*Consensus*
17b.	$(x + y)(\overline{x} + z)(y + z) = (x + y)(\overline{x} + z)$	

E.g.:

1. $A\bar{B}C+A\bar{B}\bar{C}=A\bar{B}$

   $AB\bar{C}+A\overline{B\bar{C}}=A$

2. $\bar{B}+A\bar{B}D=\bar{B}$

   $\bar{A}B+\bar{A}BCD\,(E+F)=\bar{A}B$

3. $$\begin{aligned} AB+\bar{A}C+\bar{B}C &= AB+(\bar{A}+\bar{B})\,C \\ &= AB+\overline{AB}C \\ &= AB+C \end{aligned}$$

E.g.:

- **4. a more difficult one**

$$A\overline{B}+B\overline{C}+\overline{B}C+\overline{A}B=A\overline{B}+B\overline{C}+(A+\overline{A})\,\overline{B}C+\overline{A}B\,(C+\overline{C})$$

$$=A\overline{B}+B\overline{C}+A\overline{B}C+\overline{A}\,\overline{B}C+\overline{A}BC+\overline{A}B\overline{C}$$

$$=A\overline{B}+B\overline{C}+\overline{A}C$$

Exercises:

1. $$F = A\overline{C} + ABC + AC\overline{D} + CD$$
2. $$F = AD + A\overline{D} + AB + \overline{A}C + BD + ACEF + \overline{B}EF + DEFG$$
3. $$F = BC + D + \overline{D}(\overline{B} + \overline{C})(AC + B)$$

2.7 Karnaugh Map

- Truth table in essence.
- Matrix: 2D truth table.
- Variables divided into 2 groups arbitrarily.
- Values in gray-code arrangement for every group .
- Every cell in K-map corresponds to every row in truth table.

2.7.1 Comparison between truth table and K-map

x_1	x_2	
0	0	m_0
0	1	m_1
1	0	m_2
1	1	m_3

(a) Truth table

$x_2 \backslash x_1$	0	1
0	m_0	m_2
1	m_1	m_3

(b) Karnaugh map

2.7.2 Adjacency of minterms

- For any n-variable function's minterm/maxterm, there are n adjacent minterms/maxterms.

x_2 \ x_1	0	1
0	m_0	m_2
1	m_1	m_3

2-variable K-map

x_3 \ x_1x_2	00	01	11	10
0	m_0	m_2	m_6	m_4
1	m_1	m_3	m_7	m_5

3-variable K-map

2.7.2 Adjacency of terms: continue

- For an n-variable function's minterm/maxterm, there are n adjacent minterms/maxterms.

$x_3x_4 \backslash x_1x_2$	00	01	11	10
00	m_0	m_4	m_{12}	m_8
01	m_1	m_5	m_{13}	m_9
11	m_3	m_7	m_{15}	m_{11}
10	m_2	m_6	m_{14}	m_{10}

4-variable K-map

2.7.2 Adjacency of terms: continue

- For an n-variable function's minterm/maxterm, there are n adjacent minterms/maxterms.

CD \ ABC	000	001	011	010	100	101	111	110
00	0	4	12	8	16	20	28	24
01	1	5	13	9	17	21	29	25
11	3	7	15	11	19	23	31	27
10	2	6	14	10	18	22	30	26

5-variable K-map

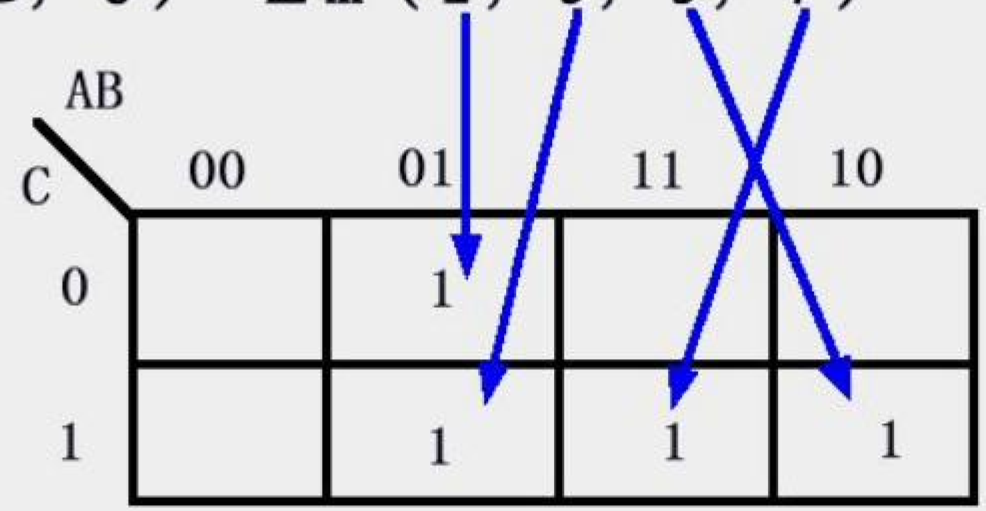
What on earth does adjacency mean?

2.7.3 Denoting a function by K-map

1. Canonical SOP

- Fill every cell with '1' corresponding to every minterm.
- E.g.:

$$F(A, B, C) = \Sigma m(2, 3, 5, 7)$$

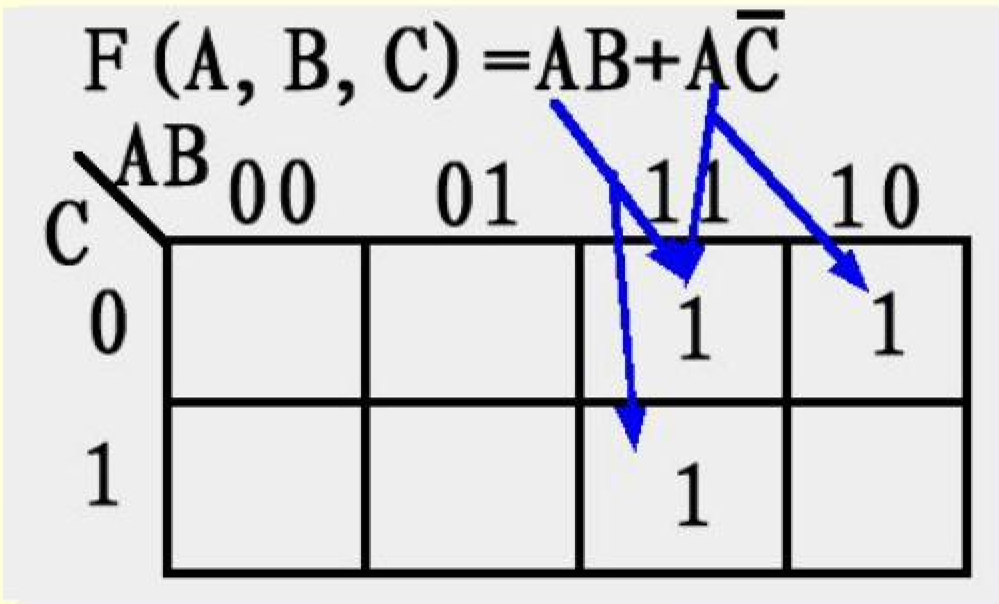


- A minterm corresponds to a cell filled with 1.

2.7.3 Denoting a function by K-map

2. General SOP

- Fill cells with ‘1s’ corresponding to every product term.
- E.g.:



- A product term corresponds to a group of cells filled with 1s.

2.7.3 Denoting a function by K-map

3. Canonical POS

- Fill every cell with ‘0’ corresponding to every maxterm.
- E.g.: $F(x_1, x_2, x_3, x_4) = \prod M(0, 1, 4, 8, 9, 12, 15)$

x_3x_4 \ x_1x_2	00	01	11	10
00	0	0	0	0
01	0	1	1	0
11	1	1	0	1
10	1	1	1	1

- A maxterm corresponds to a cell filled with 0.

2.7.3 Denoting a function by K-map

4. General POS

- Fill cells with '0s' corresponding to every sum term.
- E.g.:

$$F(x_1, x_2, x_3) = (\overline{x_1} + x_3)(\overline{x_1} + x_2)$$

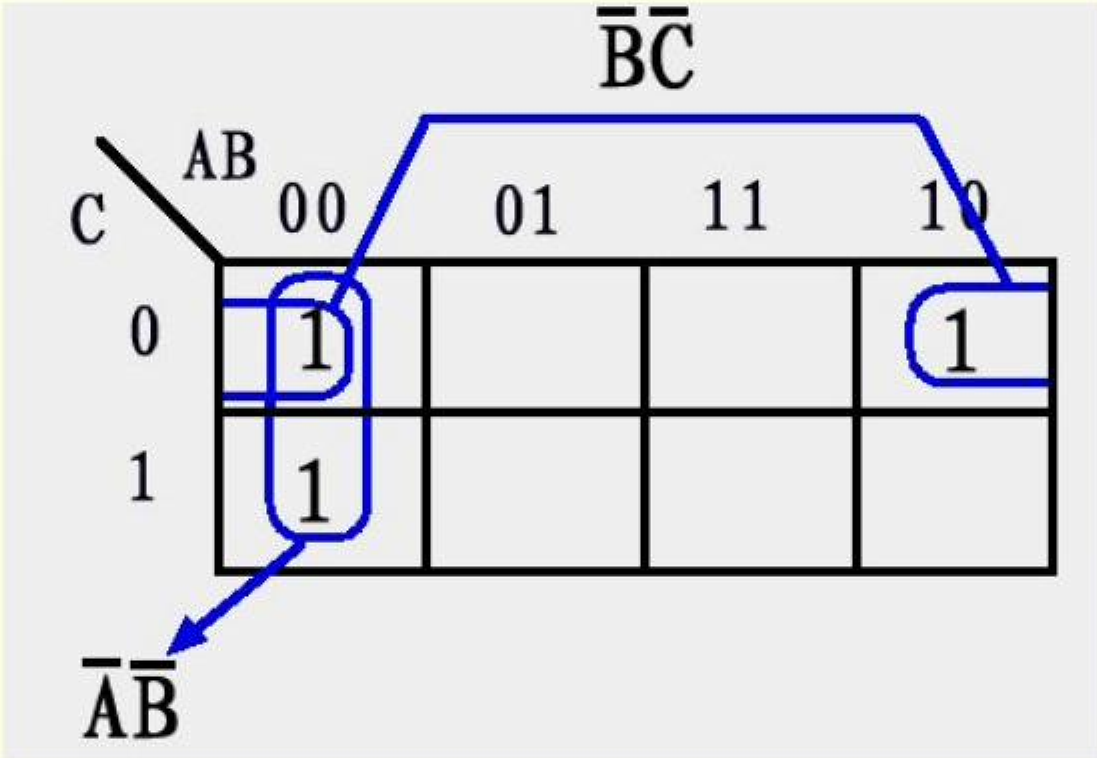
x_3 \ x_1x_2	00	01	11	10
0	1	1	0	0
1	1	1	1	0

- A sum term corresponds to a group of cells filled with 0s.

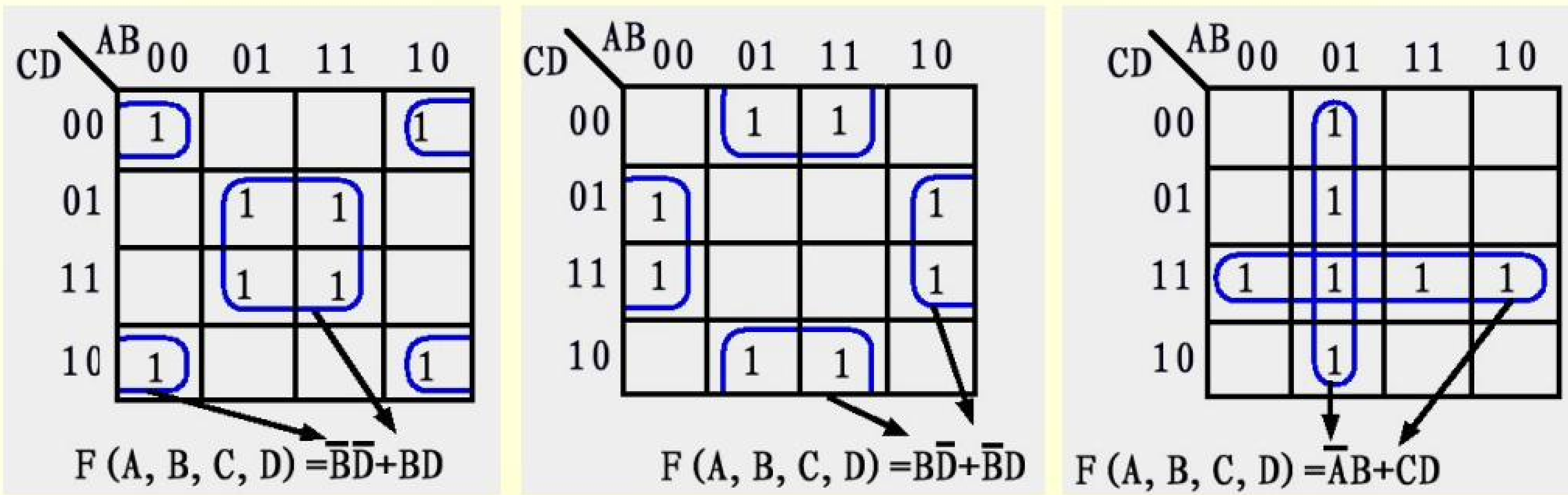
2.7.4 Key features of K-map

- **2^i adjacent 1s can be merged into a single product term, from which i variables are dropped.**
- *2^i adjacent 0s can be merged into a single sum term, from which i variables are dropped.*

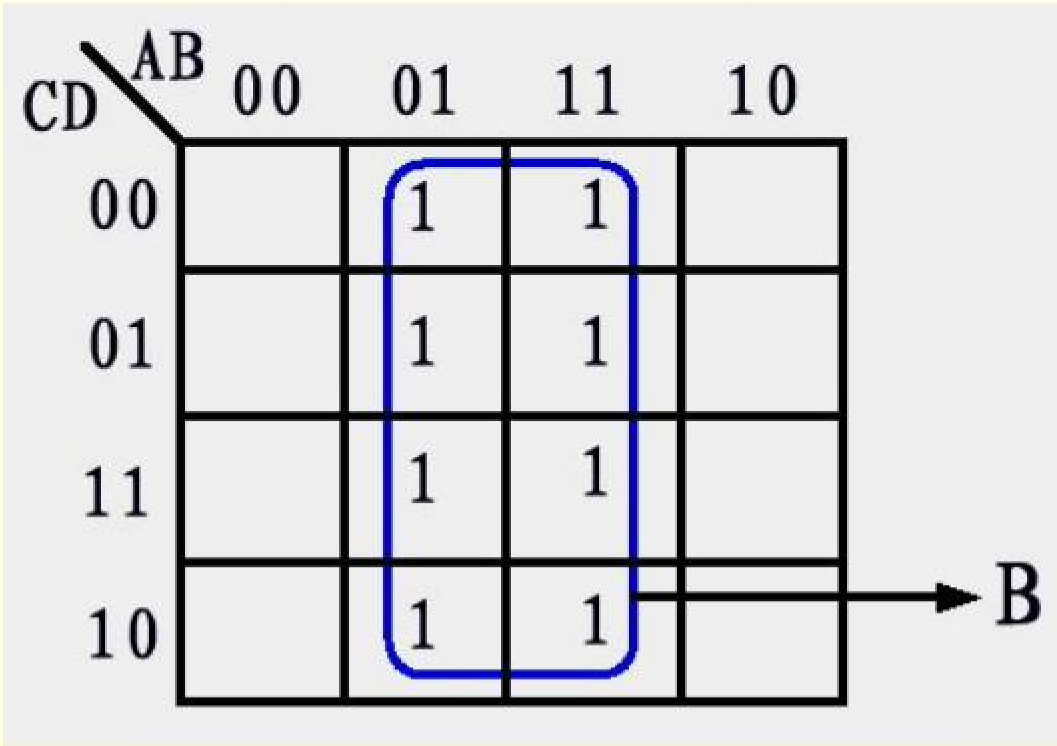
Cases of 2 adjacent 1s



Cases of 4 adjacent 1s



Cases of 8 adjacent 1s



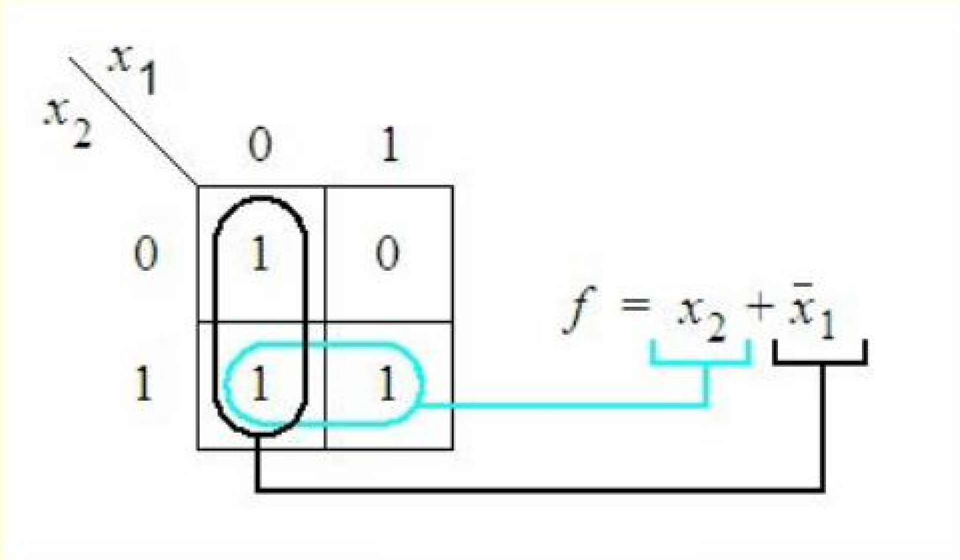
2.7.5 Strategy for Minimization

1. Finding the simplest SOP with K-map

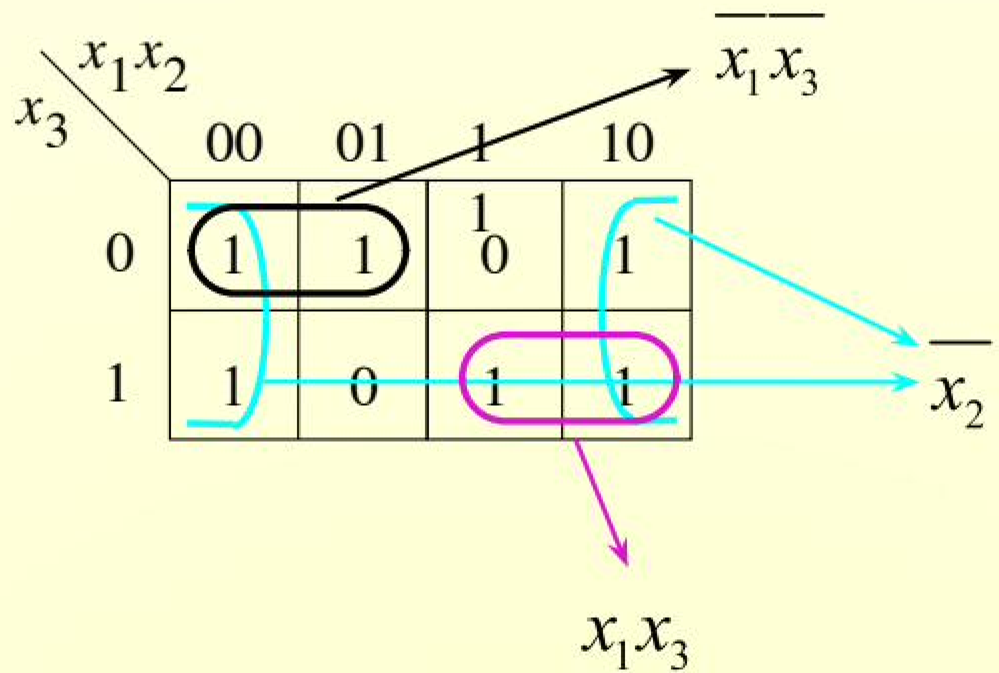
- Cover all 1s.
- 2^i adjacent 1s in every circle.
- Find as fewer groups to cover all 1s as possible(High priority)
  - *The fewer the groups ,the fewer product terms in the expression*
- Find every group(a circle) covering as many 1s as possible.
  - *The larger the group, the fewer variables in the corresponding product term.(Low priority)*

Examples

- **1.** $F(x_1, x_2) = \sum m(0,1,3)$

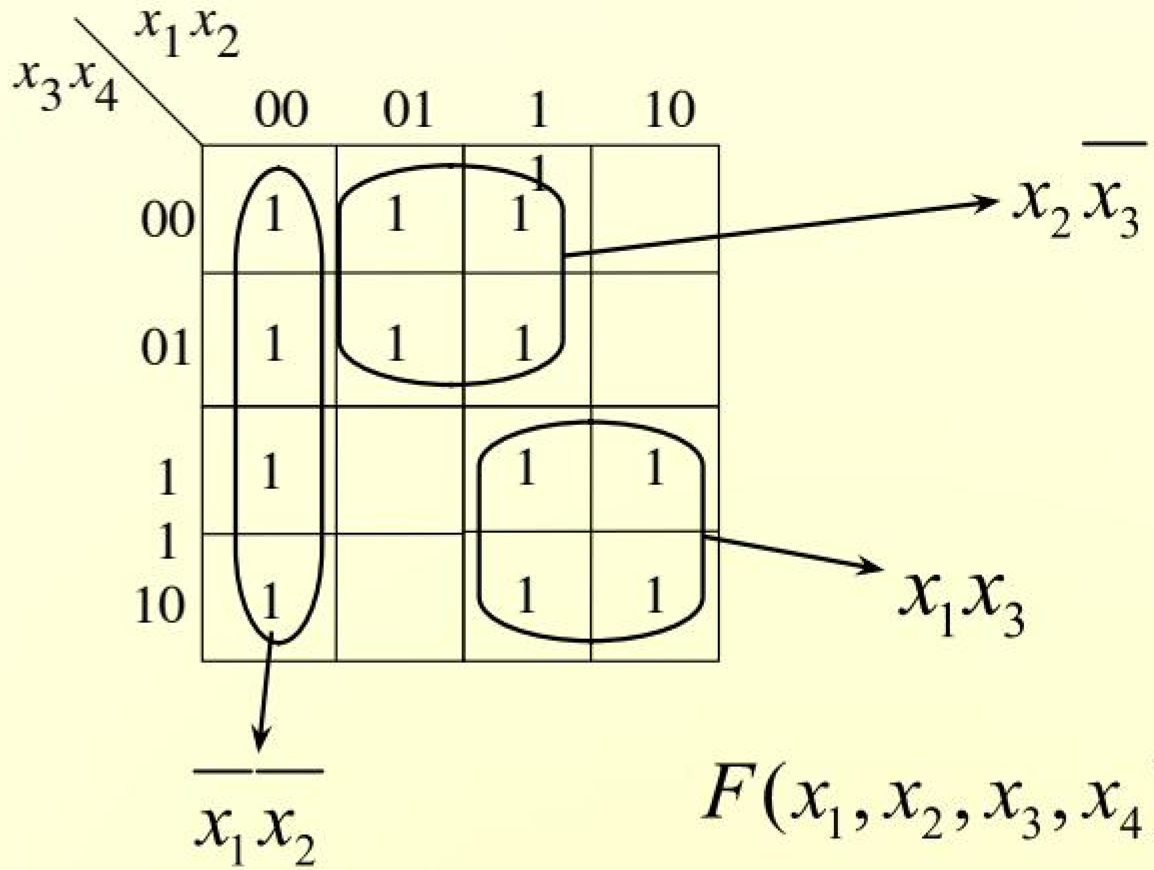


▯ **2.** $F(x_1, x_2, x_3) = \sum m(0,1,2,4,5,7)$



$$F(x_1, x_2, x_3) = \overline{x_2} + \overline{x_1}\,\overline{x_3} + x_1x_3$$

3. $F(x_1, x_2, x_3, x_4) = \overline{x_1}\,\overline{x_2} + \overline{x_1}\,\overline{x_3}x_4 + x_1x_3 + x_2\overline{x_3}$



Another solution?

$F(x_1, x_2, x_3, x_4) = \overline{x_1}\,\overline{x_2} + x_2\overline{x_3} + x_1x_3$

Exercises

$$F(A,B,C) = A\overline{B} + B\overline{C} + \overline{B}C + \overline{A}B$$

$$F(A,B,C,D) = \sum m(0,3,4,5,7,11,13,15)$$

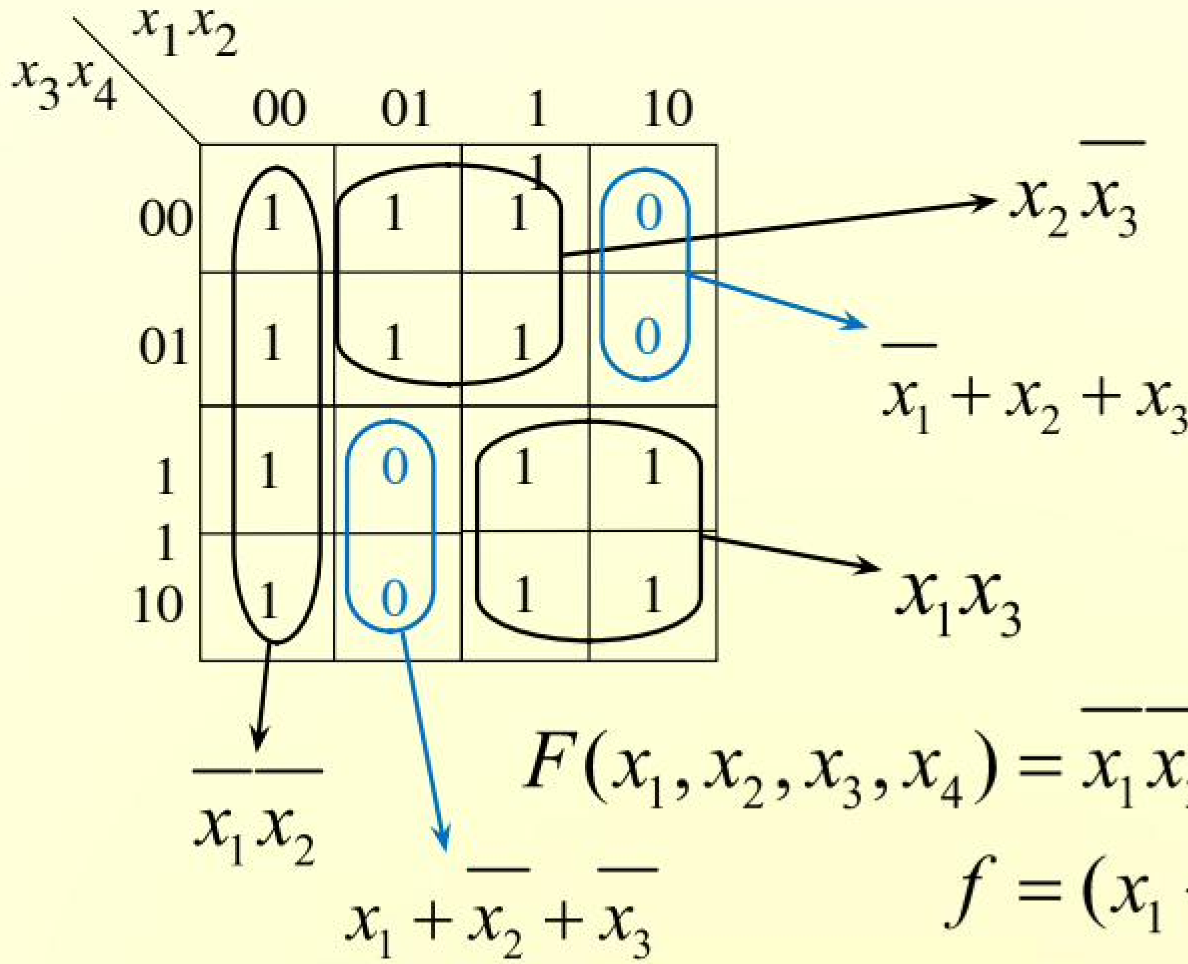
$$F(A,B,C,D) = \sum m(0,1,2,3,4,6,7,8,9,11,15)$$

2. Finding the simplest POS using K-map

- Cover all 0s.
- Cover 2^i adjacent 0s one time.
- Find every group(a circle) covering as many 0s as possible.
  - *The larger the group, the fewer variables in the corresponding sum term.*
- Find as few groups to cover all 0s as possible
  - *The fewer the groups ,the fewer sumterms in the expression.*

Examples

- 1. $F(x_1,x_2,x_3,x_4)=\overline{x_1}\,\overline{x_2}+\overline{x_1}\,\overline{x_3}x_4+x_1x_3+x_2\overline{x_3}$



$$F(x_1,x_2,x_3,x_4)=\overline{x_1}\,\overline{x_2}+x_2\overline{x_3}+x_1x_3$$

$$f=(x_1+\overline{x_2}+\overline{x_3})(\overline{x_1}+x_2+x_3)$$

Exercises

$$F(A,B,C) = A\overline{B} + B\overline{C} + \overline{B}C + \overline{A}B$$

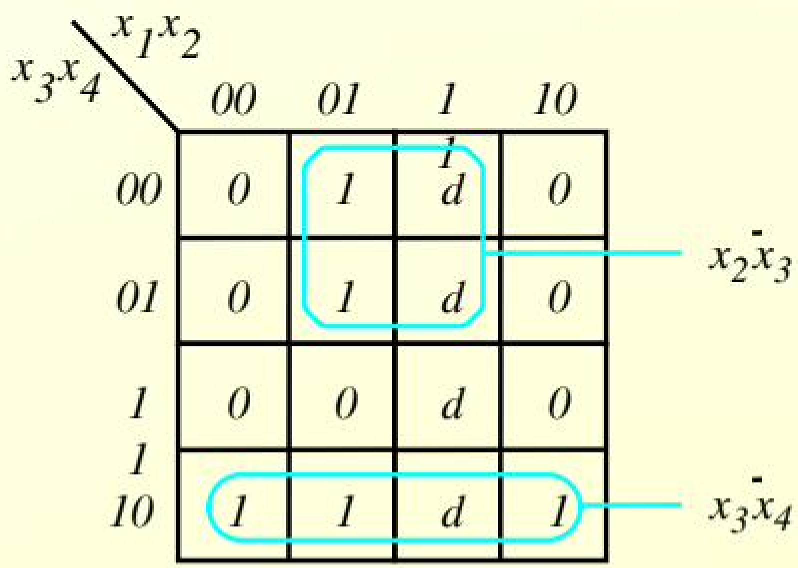
$$F(A,B,C,D) = \sum m(0,1,2,3,6,7,8,9,11,15)$$

$$F(A,B,C,D) = \prod M(2,3,4,5,6,7,11,15)$$

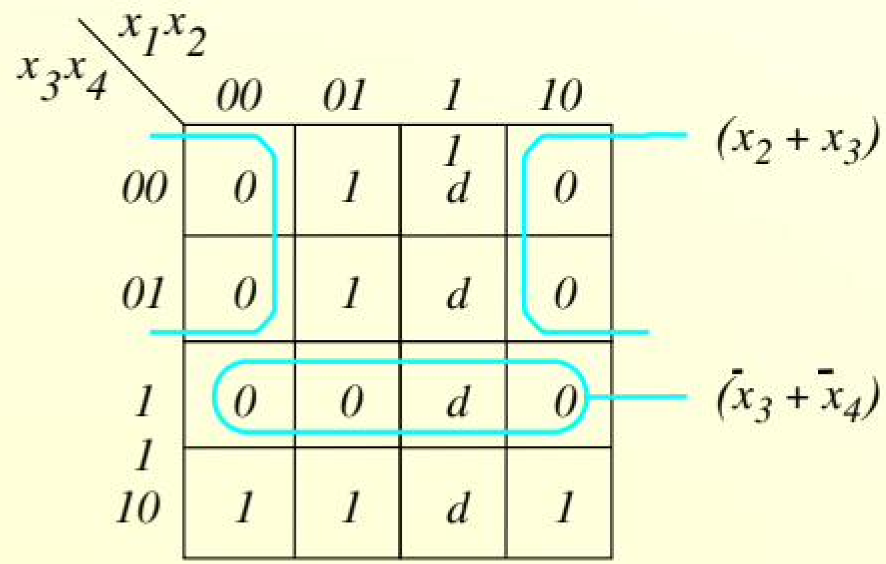
2.8 Incompletely Specified functions

- Correspond to circuits where certain input values never occur , which is called ***don't care*** condition.
- Every corresponding cell is filled with a 'd'***(don't care-term)***
- Strategy for minimization:
  - Make full use of *d*s as 1s when minimizing f into SOP.
  - Make full use of *d*s as 0s when minimizing f into POS.

E.g.:



(a) SOP implementation



(b) POS implementation

Two implementations of

$$f(x_1,\ldots,x_4) = \Sigma\, m(2, 4, 5, 6, 10) + d(12, 13, 14, 15).$$

Exercises: find the simplest SOP & POS

$$F(A,B,C,D)=\sum m(0,2,7,13,15)+\sum d(1,3,4,5,6,8,10)$$

$$F(A,B,C,D)=\sum m(0,2,3,5,7,8,10,11)+\sum d(14,15)$$

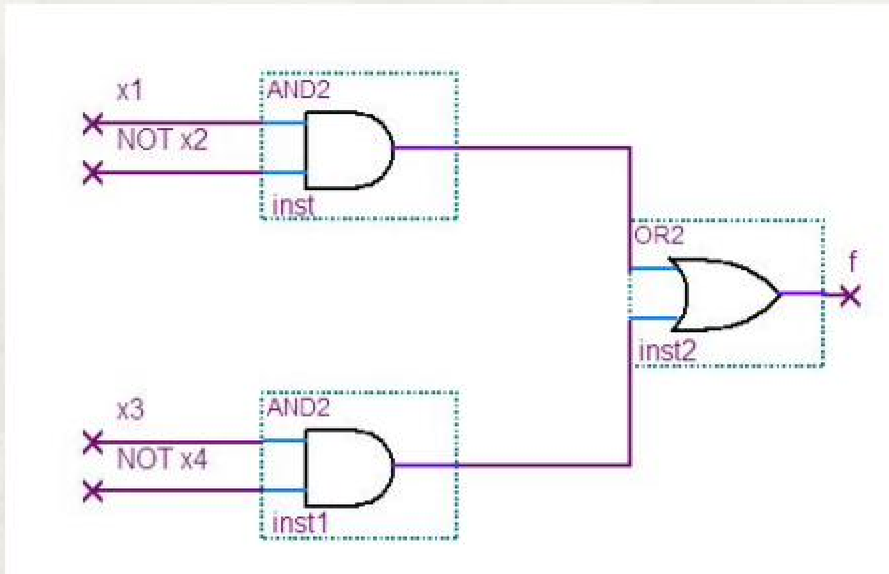
2.9 Cost

- A Function has different implementations(SOP, POS…).
- Which one entails the minimum cost?
- What is the cost?
  - Number of gates plus the total number of inputs(wires) to all gates.
  - Assumption: Single variable in true and complemented forms is at equal cost.

E.g. of cost calculation

Count the cost of $f = x_1\overline{x_2} + x_3\overline{x_4}$

1. Draw the circuit diagram:



2. 3 gates plus 6 inputs(wires) to all gates entails cost of 9

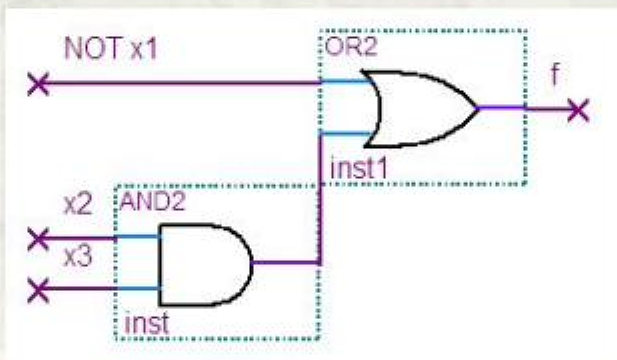
E.g. of cost calculation (cont.):
Minimum cost SOP or POS?

- A K-map

x_3 \ x_1x_2	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$\bar{x}_1$ x_2x_3

$$Min-\cos t\ \ SOP: F(x_1, x_2, x_3, x_4) = \overline{x_1} + x_2x_3$$



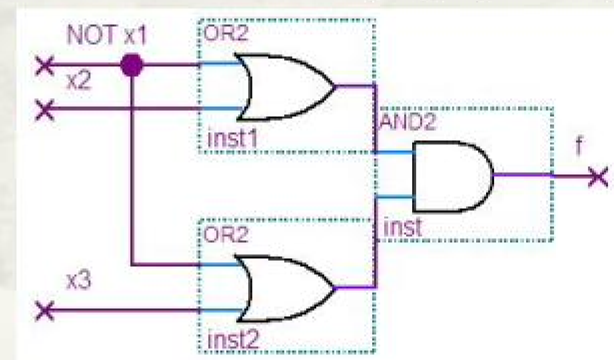
2 gates + 4 wires = 6

x_3 \ x_1x_2	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$(\bar{x}_1 + x_3)$

$(\bar{x}_1 + x_2)$

$$Min-\cos t\ \ POS: F(x_1, x_2, x_3) = (\overline{x_1} + x_2)(\overline{x_1} + x_3)$$

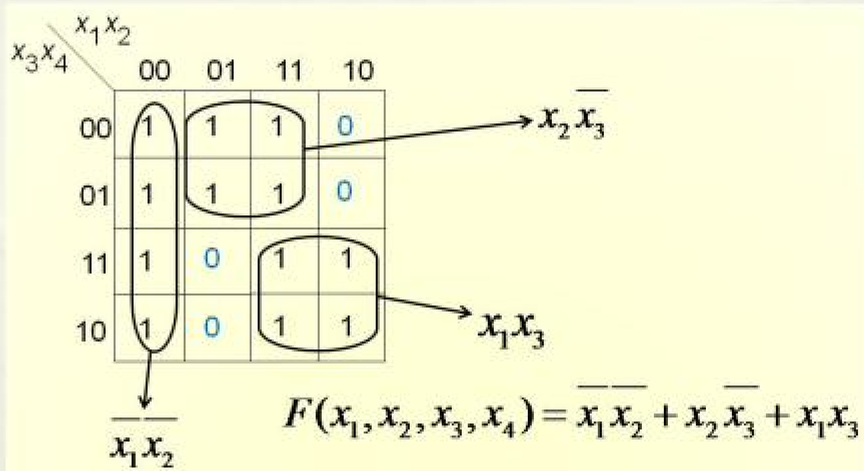


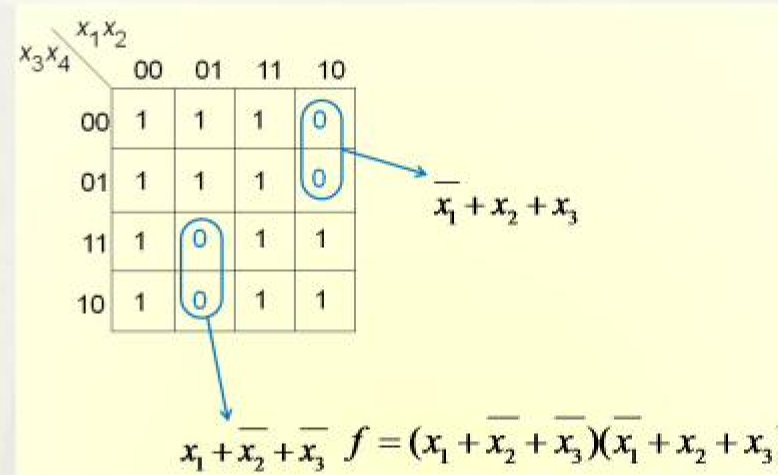
3 gates + 6 wires = 9

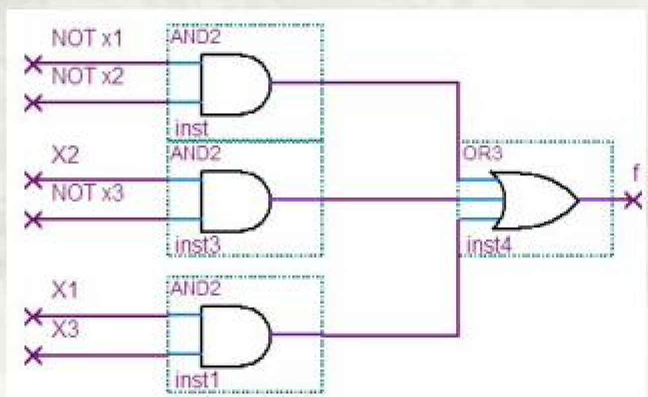
Which one is better?

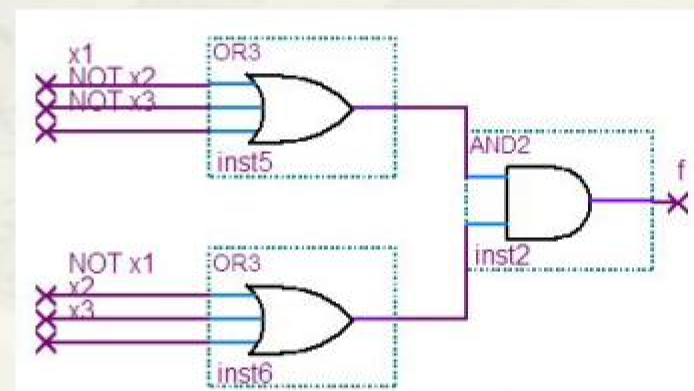
E.g. of cost calculation (cont.):
Minimum cost SOP or POS?

- A previous function: $F(x_1, x_2, x_3, x_4) = \overline{x}_1\overline{x}_2 + \overline{x}_1\overline{x}_3 x_4 + x_1x_3 + x_2\overline{x}_3$









4 gates + 9 wires = 13

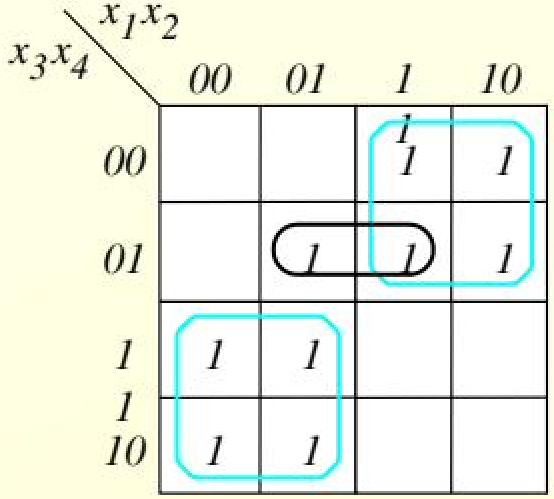
3 gates + 8 wires = 11

Which one is better?

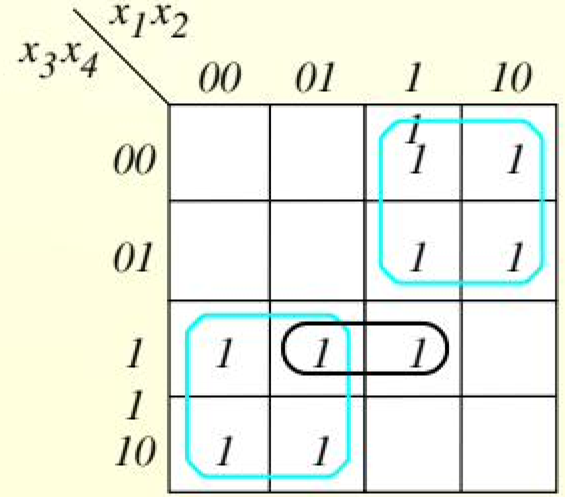
2.10 Multiple-Output functions

- A group of functions with sharing variables correspond to a multiple-output circuit.
- The whole system is not usually at minimum cost if simplifying each function separately.

E.g. 1:



(a) Function f_1



(b) Function f_2

$$f_1 = x_1\overline{x_3} + \overline{x_1}x_3 + x_2\overline{x_3}x_4$$

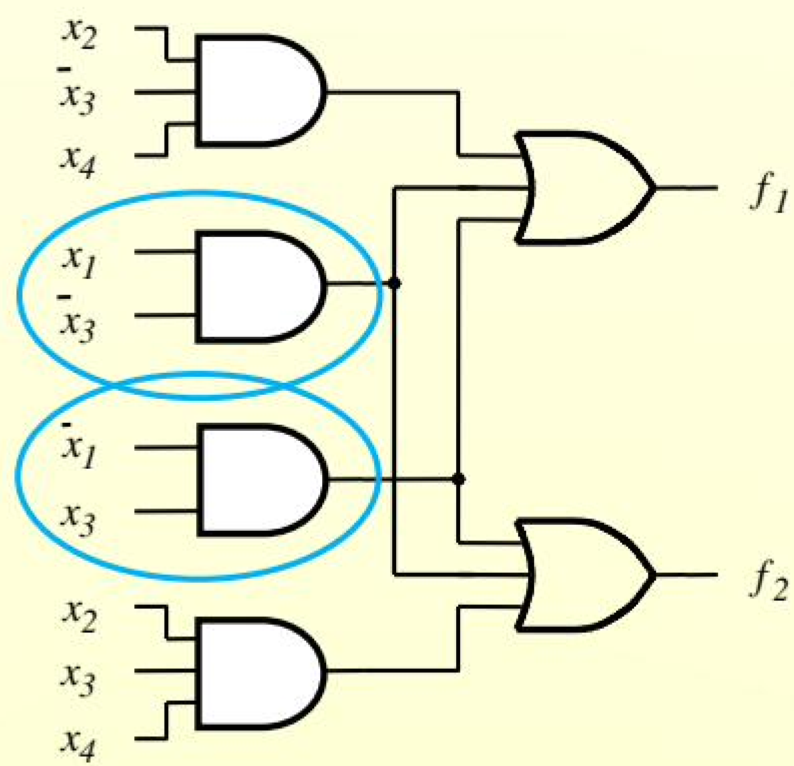
$$f_2 = x_1\overline{x_3} + \overline{x_1}x_3 + x_2x_3x_4$$

In separate design:

Total cost:14+14=28

Consider sharing product terms in *f1* and *f2*

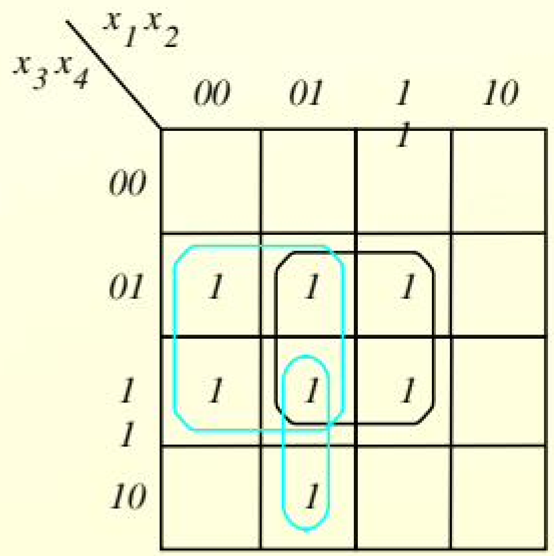
$$f_1 = \underline{x_1\overline{x}_3 + \overline{x}_1 x_3} + x_2\overline{x}_3 x_4 \qquad f_2 = \underline{x_1\overline{x}_3 + \overline{x}_1 x_3} + x_2 x_3 x_4$$



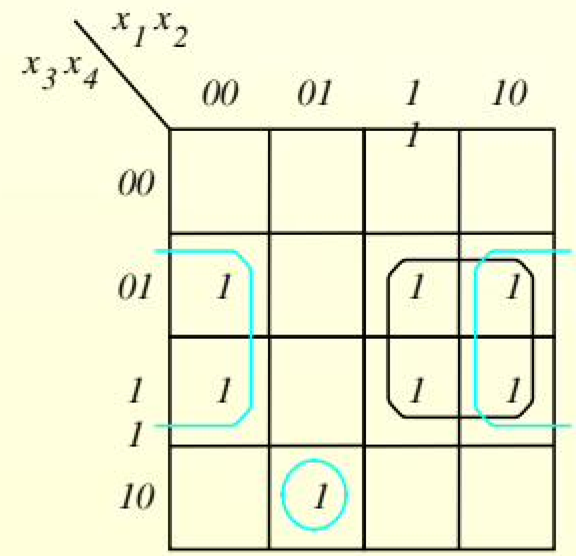
(c) Combined circuit for f_1 and f_2

Integrated cost: 28-6=22

E.g. 2:



(a) Optimal realization of f_3

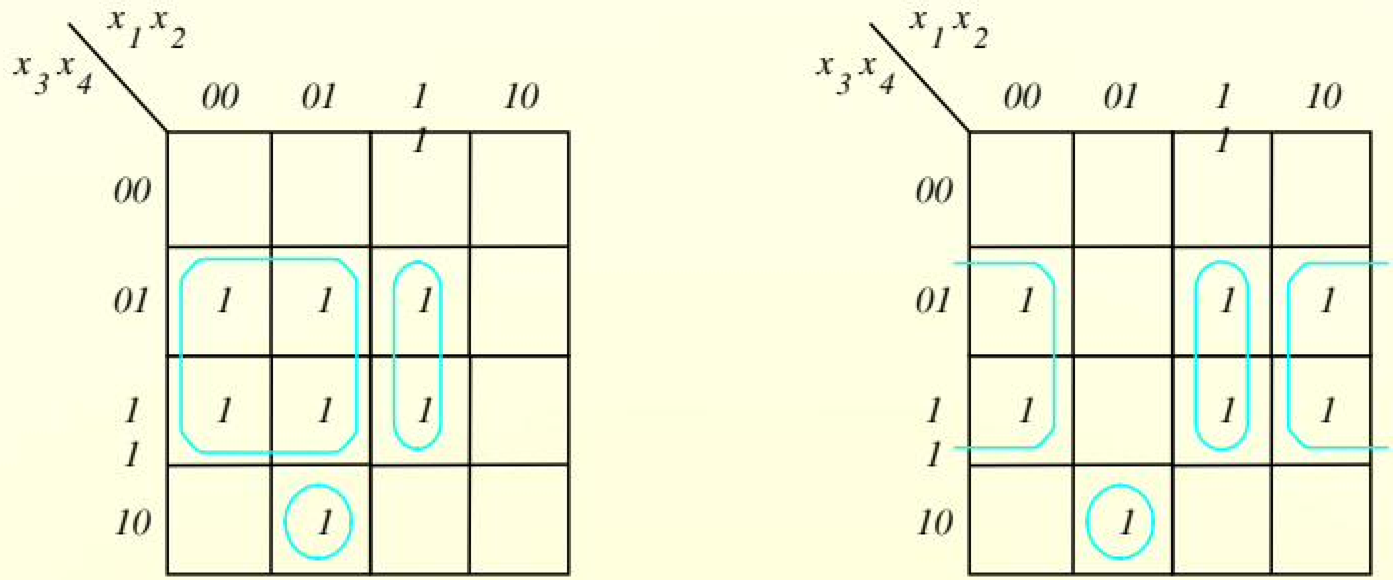


(b) Optimal realization of f_4

$$f_3 = \overline{x}_1 x_4 + x_2 x_4 + \overline{x}_1 x_2 x_3$$

$$f_4 = x_1 x_4 + \overline{x}_2 x_4 + \overline{x}_1 x_2 x_3 \overline{x}_4$$

No sharing terms!
Cost:14+15=29



(c) Optimal realization of f_3 and f_4 together

Search as many sharing terms!

$$f_3 = \overline{x}_1 x_4 + x_1 x_2 x_4 + \overline{x}_1 x_2 x_3 \overline{x}_4$$

$$f_4 = \overline{x}_2 x_4 + x_1 x_2 x_4 + \overline{x}_1 x_2 x_3 \overline{x}_4$$

Cost: <u>11+6</u>+6=23

2.11 Concluding Remarks

- Theorems in Boolean Algebra
- Manipulation of logic functions
  - Canonical SOP & POS
  - Simplest SOP & POS
- K-map
  - Structure & features?
  - How to build it given an arbitrary function?
  - Principles of simplification?
  - Special cases…